

Md Kaidul Islam

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EEE graduate from BUET with research experience in deep learning, generalized deepfake detection, and tokenization and forgetting in sequence models. Skilled in reading and critically evaluating research papers, designing rigorous experiments, and communicating technical work with clarity. Eager to build on this foundation toward NLP and agentic pipelines.

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

June 2026

B.Sc. in Electrical and Electronic Engineering

Relevant Coursework: Microwave Engineering, Communication Systems, Digital Signal Processing, Control Systems, Power Systems, Artificial Intelligence & Machine Learning, Probability & Statistics, Numerical Methods

RESEARCH INTERESTS

LLMs | Computer Vision in Medical Imaging | Forensic AI | Deep Learning | Analog & RF Circuit Design

PUBLICATION

CAE-Net: Generalized deepfake image detection using convolution and attention mechanisms with spatial and frequency domain features 

05/25

Co-author (2nd) • *Journal of Visual Communication and Image Representation (JVCIR)*, Elsevier

RESEARCH EXPERIENCE

Thesis: An Empirical Investigation of Tokenization and Forgetting in Bangla OCR

April 2025 - April 2026

Supervised by Dr. Maruf Ahmed, Associate Professor, Dept. of EEE, BUET

- Developed and evaluated PATHOK, a TrOCR-based Bangla handwritten text recognition system, comparing ViT and CNN encoders with character, grapheme, and BPE-based tokenization
- Investigated writer-level machine unlearning using writer-disjoint splits and prune-then-unlearn methods, evaluated through membership inference, linear probing, and retain-replay relearning attacks.

SELECTED PROJECTS

Hybrid Router for Token Optimization (Ongoing)

2026

    

- LLM routing system with grammar-constrained intent classification that sends complex queries to remote LLMs and use a local model for simpler tasks.

Edge-Deployable Knee MRI Abnormality Classifier (Ongoing)

2026

   

- Built a model to classify 12 knee MRI abnormalities, designed for edge-device deployment.
- MRI labels were generated using an intent classifier and API.

Smart Traffic Management via SAHI-Enhanced YOLOv8 & ByteTrack

2026

   

- Built a traffic monitoring pipeline that tiles high-resolution frames to detect smaller objects with YOLO
- SAHI tiling improved detection mAP@0.5 over baseline YOLOv8m

ESP32-Based Cold Chain Monitoring System

2025

ESP32 / Embedded Systems GSM / GPS IoT Web development

- Built an IoT monitor for medical logistics with real-time data streaming and a web dashboard.

CPLD-Based Digital Signage System

2024

Verilog Digital Design

- Designed a programmable scrolling-display signage system on an LP-2900 CPLD.

Robotic Librarian Arm

2024

Embedded Systems Automation Electromechanical Design

- Built an automated robotic arm for object handling and precise positioning.

Wilkinson Power Divider: Equal & Unequal 2-Way Power Division

2026

CST Studio Suite Microstrip / RF Circuit Design S-Parameter Analysis

- Designed and simulated equal-split and unequal-split (2:1) Wilkinson power dividers at 2.2 GHz on FR-4 and Rogers-type microstrip substrates.

TECHNICAL SKILLS

Programming & Scripting Python C / C++ MATLAB Verilog

ML / AI Frameworks PyTorch TensorFlow Keras scikit-learn Hugging Face Unisloth

Data & Vision Pandas NumPy Matplotlib OpenCV

Circuit Design & Simulation Circuit Simulation Analog Filter Design PCB Design (Proteus) CST Studio Suite

Platforms Git Kaggle Google Colab LaTeX REST APIs

Languages Bengali (native) English (proficient)

TEACHING & OUTREACH

Thakurgaon Science Society (Co-founder)

June 2025 - Present

- Co-founded an initiative to promote science education in remote and underserved communities of Bangladesh through school talks, public lectures, and hands-on science sessions.

IEEE EDS BUET Student Branch (General Secretary)

Mar 2024 - Present

- Organized seminars, workshops, and electronics education initiatives for engineering students at BUET.

BUET Literature Club (Vice President)

Mar 2025 - July 2026

- Led the organization of "BUET Boimela," a major campus literary event with coordination across external publishers and student volunteers.

SOFT SKILLS

- Published magazine editor; writer and editor of a published audiobook.
- Experienced in video editing and multimedia content production.

REFERENCES

Dr. Maruf Ahmed

Associate Professor, Dept. of EEE, BUET
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Dr. Hafiz Imtiaz

Professor, Dept. of EEE, BUET
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